

Putnam - Solutions

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Chapter 1

Putnam 1938

A1

Let $A(k) = ak^3 + bk^2 + ck + d$, where $d = M$ since $A(0) = M$. Let us compute the volume by integrating $A(k)$:

$$\begin{aligned} V &= \int_{-\frac{h}{2}}^{\frac{h}{2}} A(k) dk \\ &= \left[\frac{a}{4}k^4 + \frac{b}{3}k^3 + \frac{c}{2}k^2 + Mk \right]_{-\frac{h}{2}}^{\frac{h}{2}} \\ &= \frac{bh^3}{12} + Mh \end{aligned}$$

Our goal is to get

$$V = \frac{h(B + 4M + T)}{6}$$

Notice that

$$\begin{aligned} B &= A\left(-\frac{h}{2}\right) = -a\frac{h^3}{8} + b\frac{h^2}{4} - c\frac{h}{2} + M \\ T &= A\left(\frac{h}{2}\right) = a\frac{h^3}{8} + b\frac{h^2}{4} + c\frac{h}{2} + M \end{aligned}$$

thus

$$B + T = \frac{bh^2}{2} + 2M$$

Substituting into our goal, we simplify

$$V = \frac{h(B + 4M + T)}{6} = \frac{h(\frac{bh^2}{2} + 6M)}{6} = \frac{bh^3}{12} + Mh$$

Hence, the volume formula is indeed correct.

We can use this formula to derive the formula for a cone and sphere. For a cone with base radius r and height h , the area formula is

$$A(k) = \pi \left(\frac{r}{h} \left(k + \frac{h}{2} \right) \right)^2$$

thus a 2nd degree polynomial, which satisfies the requirements for using the special formula, hence

$$\begin{aligned} V &= \frac{h(B + 4M + T)}{6} \\ &= \frac{h(\pi r^2 + 4\pi \left(\frac{r}{2}\right)^2 + 0)}{6} \\ &= \frac{h(2\pi r^2)}{6} = \frac{\pi r^2 h}{3} \end{aligned}$$

For a sphere with radius $r = \frac{h}{2}$, the cross-sectional area is

$$A(k) = \pi \left(\frac{h^2}{4} - k^2 \right)$$

thus again a 2nd degree polynomial in k , so we apply the special formula to get

$$\begin{aligned} V &= \frac{h(B + 4M + T)}{6} \\ &= \frac{h(0 + 4\pi \frac{h^2}{4} + 0)}{6} = \frac{\pi h^3}{6} \end{aligned}$$

Expressed in radius $2r = h$, we get

$$V = \frac{4}{3}\pi r^3$$

A2

$$V = 2 \cdot \frac{\pi r^2 h}{3} + \pi r^2 h = \frac{5}{3}\pi r^2 h$$

$$A = 2\pi r h + 2\pi r \sqrt{r^2 + h^2}$$

We want to express either r or h in terms of A and the other:

$$A - 2\pi r h = 2\pi r \sqrt{r^2 + h^2}$$

$$A^2 - 4A\pi r h + 4\pi^2 r^2 h^2 = 4\pi^2 r^2 (r^2 + h^2)$$

$$A^2 - 4A\pi r h + 4\pi^2 r^2 h^2 = 4\pi^2 r^4 + 4\pi^2 r^2 h^2$$

$$A^2 - 4A\pi r h = 4\pi^2 r^4$$

$$h = \frac{A^2 - 4\pi^2 r^4}{4A\pi r}$$

Substituting into the volume equation, we get

$$\begin{aligned} V &= \frac{5}{3}\pi r^2 \frac{A^2 - 4\pi^2 r^4}{4A\pi r} \\ &= \frac{5}{12} \frac{r(A^2 - 4\pi^2 r^4)}{A} \end{aligned}$$

We want to maximize this equation given a fixed A , so we can ignore the coefficient and denominator, leaving only

$$r(A^2 - 4\pi^2 r^4) = A^2 r - 4\pi^2 r^5$$

Setting the derivative to zero, we get

$$\begin{aligned} A^2 - 20\pi^2 r^4 &= 0 \\ \Rightarrow r &= \sqrt{\frac{A}{\sqrt{20}\pi}} \end{aligned}$$

Note that, from the height equation, we know that

$$\begin{aligned} 0 &< A^2 - 4\pi^2 r^4 \\ \Rightarrow r &< \sqrt{\frac{A}{2\pi}} \end{aligned}$$

$\sqrt{20} > \sqrt{16} = 4 > 2$, so the r we found is safely within the bounds.

A3

Velocities are

$$\begin{aligned} \frac{dx}{dt} &= 3t^2 - 1 \\ \frac{dy}{dt} &= 4t^3 + 1 \end{aligned}$$

I think they are asking for the speed, which is

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{16t^6 + 9t^4 + 8t^3 - 6t^2 + 2}$$

Since we are finding the maximum, we can ignore the square root. Since a polynomial is differentiable everywhere, its maxima happen when derivative is zero

$$96t^5 + 36t^3 + 24t^2 - 12t = 0$$

Clearly, $t = 0$ is an extrema (or saddle). To know whether it's a maximum, we can take the second derivative at 0

$$480t^4 + 108t^2 + 48t - 12$$

which equals -12 (negative) when t are replaced with 0, hence $t = 0$ is a local maximum.

To know whether it's an inflection point, we can compute $\frac{d^2y}{dx^2}$ at $t = 0$, which should be possible since the tangent is not vertical (specifically, $\vec{v}(0) = (-1, 1)$):

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2 - 1}{4t^3 + 1} \\ \frac{d}{dx} \frac{dy}{dx} &= \frac{d}{dt} \frac{dy}{dx} \cdot \frac{dt}{dx} = \frac{d}{dt} \frac{dy}{dx} \cdot \frac{1}{\frac{dx}{dt}} \\ &= \frac{(4t^3 + 1)(6t) - (3t^2 - 1)(12t^2)}{(4t^3 + 1)^2} \cdot \frac{1}{3t^2 - 1} \\ &= \frac{-12t^4 + 12t^2 + 6t}{(4t^3 + 1)^2(3t^2 - 1)} \end{aligned}$$

Substituting 0, the whole thing becomes 0. Hence, the second derivative is zero, but we still need to know whether there is a sign change for the second derivative (which represents change in concavity).

For t near 0, the denominator is always negative. However, the derivative of the numerator is 6 (positive) at $t = 0$, and since the value of the numerator is 0 at $t = 0$, the numerator changes sign when $t < 0$ becomes $t > 0$. Hence, there is a sign change in the second derivative at $t = 0$, hence it's an inflection point.